isthmus-ablation-core Manual

Voxel ablation and coupling core

June 10, 2026

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`isthmus-ablation-core` is a C++ voxel ablation and coupling core. It owns solid voxel state, tracks mass and material properties, supports standalone verification runs, and is being shaped so DSMC/SPARTA can call it through thin commands, fixes, and computes.

The project requires ISTHMUS. ISTHMUS provides surface reconstruction, surface-to-voxel mapping, and the shared TIFF import utility. The core is still more than an ISTHMUS wrapper: it owns voxel bookkeeping, local voxel-face ablation, generated geometries, exact-solution verification, and DSMC coupling state.

1.1 Quick Start

If DSMC and ISTHMUS are already installed, set their roots once:

```
export DSMC_ROOT=$HOME/dsmc
export ISTHMUS_ROOT=$HOME/isthmus
```

Then build a private DSMC/IAC executable from this repository:

```
cmake --preset dsmc
cmake --build --preset dsmc
ctest --preset dsmc
```

The executable is:

```
build-dsmc/bin/dsmc-iac
```

This does not edit the DSMC checkout. The build creates a disposable overlay in `build-dsmc/dsmc-overlay/`, where DSMC source files and IAC bridge commands are symlinked together for compilation.

Run a coupled example:

```
build-dsmc/bin/dsmc-iac \
  -in examples/dsmc-sphere-kinetic/in.dsmc-sphere-kinetic
```

For standalone-only work:

```
cmake --preset standalone
cmake --build --preset standalone
ctest --preset standalone
```

1.2 Current Capabilities

- Create a slab voxel model.
- Import a small ISTHMUS-backed TIFF stack as voxels.
- Assign material density.
- Define a constant mass-loss source.
- Use explicit or mass-Courant timesteps.
- Apply local voxel ablation with delete-empty behavior.
- Use simple standalone loops with `variable`, `label`, `next`, and `jump`.

- Print aligned standalone stats with `stats` and `stats_style`.
- Write a CSV history file.
- Write VTU voxel files for visual inspection.
- Compute single-species ideal-gas kinetic-theory flux on ISTHMUS triangles.
- Read DSMC `nflux_incident` surface tallies and convert them to voxel mass

loss through ISTHMUS triangle ownership.

- Build a private DSMC executable that calls the core through thin

command-style bridge files without modifying the DSMC checkout.

- Include shared input files with `include`.
- Verify tracked quantities against input-file exact expressions.
- Optionally build visual verification reports from test data.

1.3 Rebuild Docs

Documentation source lives in `docs/`. After changing command syntax, examples, concepts, or behavior, rebuild the single PDF manual:

```
cmake -S . -B build
cmake --build build --target docs-pdf
```

The generated manual is:

```
docs/isthmus-ablation-core-manual.pdf
```

You can also call the local PDF builder directly:

```
python3 tools/build-docs-pdf.py
```

1.4 Current Run Type

The standalone executable currently prints:

```
# Standalone voxel ablation
```

Future ISTHMUS-backed runs should use a more specific title, such as:

```
# Coupled voxel/ISTHMUS surface ablation
```

1.5 First Example

```
units si

voxel_material carbon density 1800.0
voxel_create solid slab nx 8 ny 4 nz 4 dx 1.0e-6 material carbon

source q1 constant 1.8 units kg/m2/s
timestep mass/courant 0.5 source q1
variable ablation_time equal 4.0e-3

stats 1
stats_style step time active-voxels deleted-voxels remaining-mass mass-fraction volume-fraction front

voxel_dump hist solid history 1 output/slab-direct-ablation/history.csv
voxel_dump vox solid vtu 1 output/slab-direct-ablation/voxels_*.vtu select active scalar mass-fraction

label ablate-loop
limit time ${ablation_time}
voxel_ablate solid source q1 policy local face xlo delete yes
run 1
jump SELF ablate-loop until time ${ablation_time}
```

The regression wrapper in `tests/inputs/slab-direct-ablation/in.slab-direct-command.verify` includes this example and adds exact-solution checks with `verify`.

2 Getting Started

The easiest DSMC/IAC workflow is:

- Install or build DSMC and ISTHMUS somewhere on your machine.
- Tell this repository where they are.
- Build a private DSMC executable inside this repository's build directory.

Your DSMC checkout is not modified. The build creates an overlay under `build-dsmc/dsmc-overlay/` and writes the executable to:

```
build-dsmc/bin/dsmc-iac
```

2.1 One-Time Shell Setup

If DSMC and ISTHMUS live in stable locations, put these in `~/.bashrc`, `~/.zshrc`, or your shell startup file:

```
export DSMC_ROOT=$HOME/dsmc
export ISTHMUS_ROOT=$HOME/isthmus
```

Then open a new terminal or source the file:

```
source ~/.zshrc
```

You can skip this and pass paths directly to CMake instead:

```
cmake --preset dsmc -DDSMC_ROOT=/path/to/dsmc -DISTHMUS_ROOT=/path/to/isthmus
```

2.2 Build DSMC/IAC

From a fresh clone:

```
cmake --preset dsmc
cmake --build --preset dsmc
ctest --preset dsmc
```

These preset commands require CMake 3.20 or newer. Check with:

```
cmake --version
```

If your machine has an older CMake, either load a newer module or use the non-preset form:

```
cmake -S . -B build-dsmc -DIAC_DSMC_USE_OVERLAY=ON
cmake --build build-dsmc --target dsmc
ctest --test-dir build-dsmc --output-on-failure
```

The `dsmc` build preset:

- builds `libisthmus_ablation_core.a`;
- creates `build-dsmc/dsmc-overlay/src`;

- symlinks DSMC source files into that overlay;
- symlinks the IAC bridge command files into that overlay;
- generates the overlay `Makefile.package`;
- runs DSMC's normal `make <machine>` build there;
- writes `build-dsmc/bin/dsmc-iac`.

The default DSMC machine target is `mpi`. Override it with:

```
cmake --preset dsmc -DDSMC_MACHINE=mac_mpi
```

or without presets:

```
cmake -S . -B build-dsmc -DIAC_DSMC_USE_OVERLAY=ON -DDSMC_MACHINE=mac_mpi
```

2.3 Run A DSMC/IAC Case

```
build-dsmc/bin/dsmc-iac \
  -in examples/dsmc-sphere-kinetic/in.dsmc-sphere-kinetic
```

Run the fast DSMC verification suite:

```
ctest --preset dsmc
```

2.4 Standalone Only

The standalone binary does not need DSMC:

```
cmake --preset standalone
cmake --build --preset standalone
ctest --preset standalone
```

Without presets:

```
cmake -S . -B build -DIAC_DSMC_USE_OVERLAY=OFF
cmake --build build
ctest --test-dir build --output-on-failure
```

Run the first standalone example:

```
./build/ia-core -in examples/slab-direct-ablation/in.slab-direct-ablation
```

2.5 Visual Outputs

Examples write under `output/` by default. For the direct slab example:

```
output/slab-direct-ablation/history.csv
output/slab-direct-ablation/voxels_000000.vtu
output/slab-direct-ablation/voxels_000001.vtu
...
```

Open VTU/VTP files directly in ParaView. The examples usually write active voxels only and select `mass-fraction` as the first visible cell scalar.

2.6 Rebuild The Manual

```
cmake --build --preset docs
```

The PDF is written to:

```
docs/isthmus-ablation-core-manual.pdf
```

3 Architecture

The project has two intended front doors into one shared core:

- Standalone mode: this repository owns parsing, timestep advancement, stats,

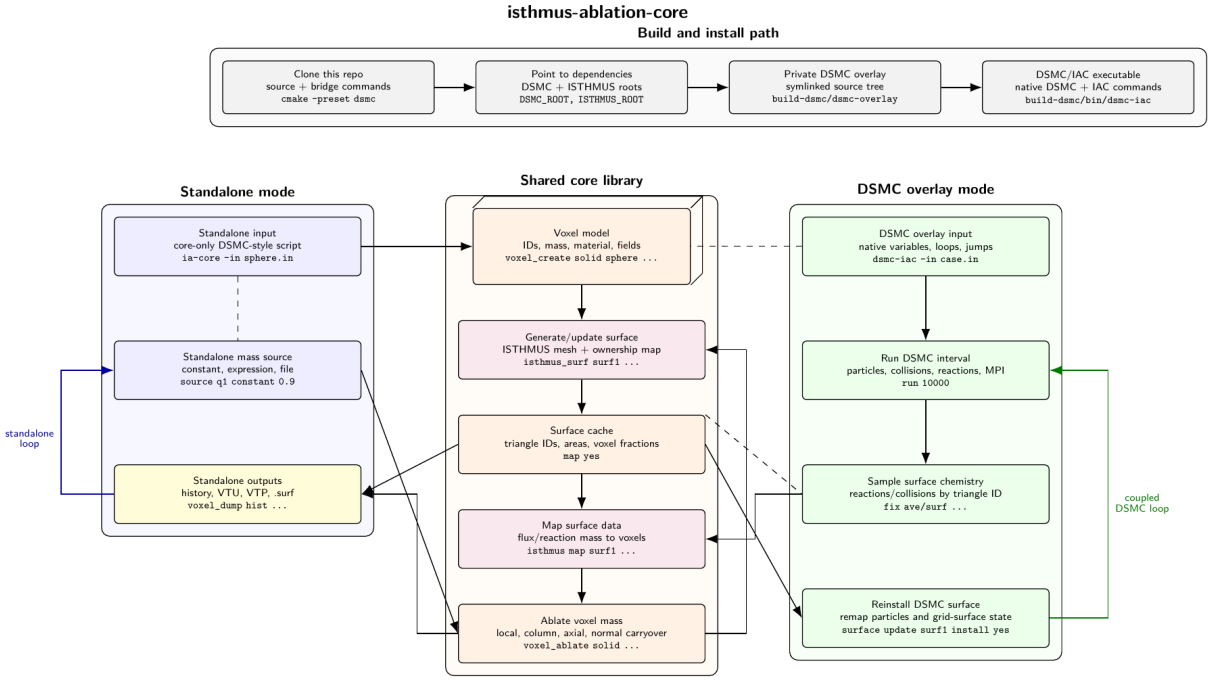
dumps, and verification.

- DSMC/SPARTA-linked mode: DSMC/SPARTA owns the main input parser and timestep

loop, while this project contributes commands, computes, and a core library object.
The shared core owns:

- voxel state with stable IDs;
- material density and voxel mass;
- generated and imported geometries;
- mass-loss sources;
- ablation policies;
- diagnostics and history;
- ISTHMUS surface state and ownership mapping.

The current flowchart is available here:



Architecture flowchart

Architecture flowchart

3.1 ISTHMUS Coupling

The current standalone ISTHMUS path is:

- Convert active voxels to an `isthmus::VoxelSet`.
- Call `isthmus::MarchingWindows`.
- Cache the surface mesh and triangle-to-voxel ownership fractions.
- Run `surf_flux ...` to apply mass flux to selected triangles.
- Run `voxel_ablate ...` to update voxel mass and delete empty voxels.
- Run `isthmus_surf ...` again if the voxel state changed.

3.2 Future DSMC/SPARTA Coupling

The planned DSMC coupling should support both a command-loop path and a timestep-callback path. We will implement whichever proves easiest first.

The command-loop path would look like:

```
label ablate-loop
run 10000
surf_flux surf1 source dsmc-flux select all
voxel_ablate solid surface surf1 policy local delete yes
isthmus_surf surf1 voxels solid map yes
surface update surf1 install yes
jump SELF ablate-loop
```

The loop keeps DSMC/SPARTA in charge of particles, reactions, variables, labels, jumps, MPI, and stats. This project supplies commands that operate on the shared voxel/surface core between DSMC run intervals.

The callback path would expose similar core operations through a fix-style or other timestep callback so geometry can update during a longer run. That may be more natural for cases that need frequent ablation updates.

The main tradeoff is timing control. Command loops are explicit and fit native DSMC/SPARTA `label`, `jump`, and chunked `run` workflows. Callback coupling can hide more machinery behind a timestep hook, but may be more convenient for fine-grained updates.

4 Verification

Verification should be input-driven, not hardcoded in the core model.

The core tracks actual quantities such as:

- `remaining-mass`
- `mass-fraction`
- `volume-fraction`
- `front`
- `active-voxels`
- `deleted-voxels`

The input file declares exact or reference expressions:

```
verify mass-fraction exact "1.0 - q1*time/(rho*length)" tolerance 0.01 percent norm max
verify volume-fraction exact "1.0 - q1*time/(rho*length)" tolerance 0.01 percent norm final
verify front exact "q1*time/rho" tolerance 0.01 percent norm final
verify radius exact "initial-radius - q1*time/rho" tolerance 3.0 percent norm final
```

This keeps geometry-specific exact solutions out of the source code.

`mass-fraction` is the smooth remaining-mass ledger normalized by initial mass. `volume-fraction` is the active voxel count normalized by the initial active voxel count. It changes only when voxels are deleted, so it captures the stair-stepped voxelized geometry response.

4.1 Norms

`norm final` checks only the last history row.

`norm max` checks the maximum absolute error over recorded history rows.

`norm rms` checks the root-mean-square error over recorded history rows.

Regression tests should prefer percent tolerances unless an absolute error scale is more meaningful.

4.2 Convergence

Convergence checks are input-driven too. A test can define variables, use them inside the case setup, and ask the executable to rerun the same input with overrides:

```
variable resolution equal 20
variable steps equal 4
voxel_create solid sphere diameter 1.0e-3 resolution ${resolution} material carbon
variable i loop ${steps}

convergence radius exact "initial-radius - q1*time/rho" tolerance 100.0 percent norm final vary resolution 5 10 20 va
convergence volume-fraction exact "((initial-radius - q1*time/rho)/initial-radius)^3" tolerance 100.0 percent norm fi
```

This keeps convergence criteria in the test input instead of a separate helper script.

5 Expected Features

This page tracks planned user-visible capabilities. It is intentionally more stable than planning notes and brainstorm documents.

5.1 Voxel Core

- Generate slab, sphere, cylinder/cone, and imported voxel geometries.
- Track stable voxel IDs, lattice coordinates, material, remaining mass,

activity, fixed state, and future fields.

- Support local ablation, column carryover, axial carryover, normal-directed

carryover, ghost voxels, and configurable boundary behavior.

- Dump history, VTU voxel state, restart state, and diagnostic summaries.

5.2 Ablation Control

The long-term design should support both explicit commands and timestep callbacks. We will implement whichever path is easiest first while keeping the core API usable by both.

The command-oriented form is useful for readable standalone scripts and DSMC/SPARTA loops:

```
label ablate-loop
voxel_ablate solid source q1 policy local face xlo delete yes
isthmus_surf surf1 voxels solid map yes
surf_flux surf1 source dsmc-tallies select all
voxel_ablate solid surface surf1 policy local delete yes
jump SELF ablate-loop
```

Standalone inputs can loop over these commands directly. DSMC/SPARTA inputs can use native variables, labels, jumps, and repeated `run` chunks to couple gas evolution and voxel/surface updates.

The callback form may use a fix-style or other timestep hook to call the same core operations during a longer DSMC/SPARTA run.

5.3 ISTHMUS Coupling

- Generate a surface from active voxels using ISTHMUS.
- Cache triangle IDs, areas, normals, and triangle-to-voxel ownership

fractions.

- Map surface quantities back to voxel mass loss.
- Regenerate the surface when voxel state changes.
- Export VTP and SPARTA `.surf` files in standalone mode.
- Install or replace explicit DSMC/SPARTA surfaces in coupled mode.

5.4 DSMC/SPARTA Coupling

- Let DSMC/SPARTA own particle advancement, collisions, reactions, MPI,

variables, loops, and stats.

- Read per-surface DSMC tallies after a `run` interval.
- Map those tallies to ISTHMUS triangles and then to voxels.
- Apply voxel mass loss with chosen policies.
- Regenerate and reinstall the surface before the next `run` interval.

6 Command Reference

The command language is intentionally close to DSMC/SPARTA style: one command per line, simple positional arguments, and keyword/value options.

New command names avoid underscores where this project controls the syntax. Existing DSMC/SPARTA commands keep their native spelling, for example `stats_style` and `surf_modify`.

Current standalone commands:

`include` (include.md)

`voxel` (voxel.md)

`source` (source.md)

`timestep` (timestep.md)

`el`, `next`, and `jump` (loops.md)

`voxel_ablate` (voxel-ablate.md)

`..` `voxel/ablate` (fix-voxel-ablate.md)

`isthmus_surf` (isthmus-surface.md)

`surface` (surface.md)

`iac` (iac.md) for DSMC-hosted diagnostics and verification

`s` and `stats_style` (stats.md)

`voxel_dump` (voxel-dump.md)

`verify` (verify.md)

`run` (run.md)

7 `include` Command

The `include` command reads another input file at the point where the command appears. This keeps examples and tests from duplicating the same setup.

7.1 Syntax

```
include <path>
```

7.2 Examples

```
include ../../../examples/slab-direct-ablation/in.slab-direct-ablation
```

7.3 Description

Relative paths are resolved from the directory of the file that contains the `include` command.

The command is useful for regression tests:

```
include ../../../examples/slab-direct-ablation/in.slab-direct-ablation

verify remaining-mass exact "initial-mass - q1*area*time" tolerance 0.01 percent norm max
verify mass-fraction exact "1.0 - q1*time/(rho*length)" tolerance 0.01 percent norm max
verify front exact "q1*time/rho" tolerance 0.01 percent norm final
```

The example owns the physical case. The test wrapper owns the pass/fail criteria.

8 `voxel` Command

The `voxel` command family creates and configures voxel-state objects.

8.1 Syntax

```
voxel_material <name> density <rho> [molar-mass <kg/mol>] [formula <formula>]
voxel_create <model> slab nx <nx> ny <ny> nz <nz> dx <dx> material <name>
voxel_create <model> sphere diameter <D> dx <dx> material <name>
voxel_create <model> sphere diameter <D> resolution <N> material <name>
voxel_create <model> tiff file <path> dx <dx> material <name> \
  [ox <x>] [oy <y>] [oz <z>]
voxel_write_history <model> <path>
```

8.2 Examples

```
voxel_material carbon density 1800.0 molar-mass 0.0120107 formula C
voxel_create solid slab nx 8 ny 4 nz 4 dx 1.0e-6 material carbon
voxel_create solid sphere diameter 1.0e-3 dx 5.0e-5 material carbon
voxel_create solid sphere diameter 1.0e-3 resolution 20 material carbon
voxel_create solid tiff file examples/tiff-carbon-sample/carbon-sample-top-crop.tif \
  dx 3.3757e-6 material carbon
voxel_ghost solid axis y boundary infinite layers 1
voxel_write_history solid output/dsmc-sphere-kinetic/history.csv
```

8.3 Description

`voxel_material` defines a material and its density. `molar-mass` and `formula` are optional for purely mechanical voxel ablation, but are used by DSMC-coupled surface-flux commands that infer solid mass consumption from a SPARTA surface-reaction file.

`voxel_create` creates a named voxel model. Each voxel receives a stable ID, integer lattice indices, a centroid, an initial mass based on density and `dx^3`, and an active flag.

`slab` fills the full rectangular lattice.

`sphere` fills voxel centroids inside a sphere with diameter `D`. Specify either `dx` or `resolution`, but not both. `resolution <N>` means `N` voxels across the sphere diameter, so `dx = D/N`. Voxels outside the sphere remain in the bounding lattice as inactive cells so the structured indexing and voxel IDs stay stable while the sphere ablates.

`tiff` imports a multipage TIFF stack through ISTHMUS's TIFF utility. The current shared utility supports a narrow uncompressed 8-bit grayscale stack convention, with voxels whose image value is `1` treated as active. The active coordinates returned by ISTHMUS are expanded into a structured IAC voxel lattice. `ox`, `oy`, and `oz` optionally shift the voxel-centroid origin.

`voxel_ghost` configures ghost voxel images used during surface generation. The first supported boundary is `boundary infinite`, which extrapolates active boundary voxels beyond both ends of the requested axis. Ghost voxels do not carry independent mass; their surface ownership maps back to the corresponding real voxel. This is useful for making a finite slab patch behave like an infinite wall during ISTHMUS marching cubes.

`voxel_write_history` writes the current in-memory history table immediately. Standalone inputs usually use `voxel_dump <id> <model> history ...`; the direct write command is mainly for DSMC-hosted scripts, where DSMC owns the loop and the core history should be flushed after the coupled loop completes.

8.4 Current Limits

- Only one voxel model is currently supported.

- Only one material is currently supported.
- Slab, centroid-filled sphere, and ISTHMUS-backed narrow 8-bit TIFF-stack

geometry are currently implemented.

- Ghost voxels currently support `boundary infinite`.

8.5 Planned Extensions

- CSV import.
- Periodic and open ghost voxel behavior.

9 `source` Command

The `source` command defines mass-loss data. It is available in standalone inputs and in DSMC-hosted inputs through the bridge.

9.1 Syntax

```
source <id> constant <value> units <units>
```

9.2 Example

```
source q1 constant 1.8 units kg/m2/s
```

9.3 Description

The current implementation supports a constant mass flux source. The value is interpreted as mass flux in `kg/m2/s` for the local slab ablation case.

In DSMC-hosted inputs, `source` is mainly useful for no-gas verification cases, debug probes, and prescribed ablation studies. Coupled DSMC chemistry cases usually use `surf_flux dsmc/reaction` or `surf_flux dsmc/surf` instead.

9.4 Current Limits

- Only `constant` sources are implemented.
- Units are recorded but not converted.

9.5 Planned Extensions

- Expressions.
- File or table sources.
- Surface-triangle sources.

10 `timestep` Command

The `timestep` command sets the standalone time increment.

10.1 Syntax

```
timestep <dt>
timestep mass/courant <C> source <source-id>
```

10.2 Examples

```
timestep 1.0e-6
timestep mass/courant 0.5 source q1
```

10.3 Description

The explicit form sets `dt` directly.

The mass-Courant form chooses:

```
dt = C * rho * dx / source
```

For the current slab case, `C = 0.5` removes half of one voxel mass from each exposed column per step.

11 Loop Commands

The standalone runner supports a small DSMC/SPARTA-style loop subset.

11.1 Syntax

```
variable <name> loop <N>
label <name>
next <name>
jump SELF <label>
jump SELF <label> until time <target>
limit time <target>
```

11.2 Example

```
variable ablation_time equal 4.0e-3
label ablate-loop
limit time ${ablation_time}
voxel_ablate solid source q1 policy local face xlo delete yes
run 1
jump SELF ablate-loop until time ${ablation_time}
```

11.3 Description

`variable <name> loop <N>` creates a loop counter.

`label <name>` marks a target location.

`next <name>` advances the loop counter. When the counter is exhausted, the next `jump` command is skipped.

`jump SELF <label>` jumps to a label in the current input.

`jump SELF <label> until time <target>` jumps only while the standalone ablation clock is below the target time. This is the preferred pattern for examples because the input states the physical ablation time directly.

`limit time <target>` clips the current timestep if the next `run` command would advance past the target. Put it inside the loop before `voxel_ablate` so the final mass-loss update and the recorded time end exactly at the target.

The current implementation supports only these compact loop patterns. It is meant to make standalone inputs resemble DSMC/SPARTA coupled scripts without reimplementing the full DSMC/SPARTA variable language.

12 `voxel_ablate` Command

The `voxel_ablate` command applies one timestep of mass loss to a voxel model.

12.1 Syntax

```
voxel_ablate <model> source <source-id> policy <policy> face <face> delete <yes|no>
voxel_ablate <model> surface <surface-id> policy <policy> delete <yes|no>
```

12.2 Example

```
voxel_ablate solid source q1 policy local face xlo delete yes
voxel_ablate solid source q1 policy local face yhi delete yes
voxel_ablate solid surface skin policy local delete yes
voxel_ablate solid surface skin policy carryover/normal delete yes
```

12.3 Description

The command applies one mass-loss update over the current timestep.

With `source`, `policy local` removes mass from the first active voxel in each slab column on the selected face. The required `face` argument accepts `xlo`, `xhi`, `ylo`, `yhi`, `zlo`, or `zhi`.

With `surface`, the command consumes mass already assigned to triangles by `surf_flux`, uses the ISTHMUS triangle-to-voxel ownership map, and subtracts that mass from the associated voxels.

`policy local` removes only the mass available in each mapped voxel and records any overshoot as dropped mass.

`policy carryover/normal` conserves overshoot by pushing excess mass loss from a depleted voxel into live 26-neighbor voxels along the inward normal direction. This policy is intended for closed ISTHMUS surfaces such as spheres.

Use `run 1` after `voxel_ablate` to advance time, record history, print stats, and write scheduled dumps:

```
voxel_ablate solid source q1 policy local face xlo delete yes
voxel_ablate solid source q1 policy local face zhi delete yes
run 1
```

A surface-coupled loop is:

```
isthmus_surf skin voxels solid buffer 1 weighting no map yes
surf_flux skin source q1 select all
voxel_ablate solid surface skin policy carryover/normal delete yes
run 1
```

`delete yes` is the default. When a voxel's mass is depleted, it is marked inactive and omitted from VTU dumps that use `select active`.

12.4 Current Limits

- Only one voxel model is currently supported.
- `policy carryover/normal` is currently implemented for surface-backed

ablation.

- The current source must be a constant mass flux.

13 `fix ... voxel/ablate` Command

The `voxel/ablate` fix applies mass loss to a voxel model during standalone time advancement. It is the compact callback-style alternative to explicit `voxel_ablate` commands.

13.1 Syntax

```
fix <id> all voxel/ablate <Nevery> voxels <model> source <source-id> policy <policy> face <face> delete <yes|no>
```

13.2 Example

```
fix ab all voxel/ablate 1 voxels solid source q1 policy local face xlo delete yes
fix ab all voxel/ablate 1 voxels solid source q1 policy local face yhi delete yes
```

13.3 Description

Every `Nevery` steps, the fix applies the selected source to the selected voxel model. The current implementation supports a local slab face update:

- find the first active voxel from the selected face in each slab column;
- remove `source * dx^2 * dt` from that voxel;
- delete the voxel if its mass reaches zero and `delete yes` is set.

The required `face` argument accepts `xlo`, `xhi`, `ylo`, `yhi`, `zlo`, or `zhi`.

`delete yes` is the default behavior. When a voxel's mass is depleted, it is marked inactive and omitted from VTU dumps that use `select active`. Use `delete no` only for cases where depleted voxels should remain in the active voxel set.

13.4 Current Limits

- Only group `all` is accepted.
- Only policy `local` is implemented.
- Only slab face ablation is implemented.

13.5 Planned Extensions

- Column carryover.
- Axial carryover.
- Normal-directed carryover.
- Shared core support for both explicit `voxel_ablate` commands and fix-style

timestep callbacks.

14 `isthmus_surf` Command

The `isthmus_surf` command reconstructs a triangle surface from active voxels.

14.1 Syntax

```
isthmus_surf <surface-id> voxels <model> [buffer <N>] [weighting yes|no] [map yes|no] [crop real|no]
```

14.2 Example

```
isthmus_surf skin voxels solid buffer 1 weighting no map yes
isthmus_surf skin voxels solid buffer 1 weighting no map yes crop real
```

14.3 Description

The command sends the active voxels in `<model>` to ISTHMUS, runs marching cubes, stores the resulting triangle mesh, and optionally stores triangle-to-voxel ownership fractions.

`map yes` is required when the surface will be used for ablation, because `voxel_ablate <model> surface <surface-id>` needs those ownership fractions to send triangle mass loss back to voxels.

`buffer` adds empty voxel layers around the active voxel bounding box before marching cubes. This helps ISTHMUS generate a closed surface as the solid shrinks.

`crop real` removes triangles whose centroids are outside the real voxel domain after ghost voxels are added. Use it when ghost voxels are only a boundary condition for surface quality and flux should still be applied only on the real DSMC/voxel domain.

14.4 DSMC Coupling Note

This command is intentionally an explicit input-file command rather than a hidden fix callback. A future DSMC loop can call it after voxel mass changes and before refreshing the DSMC surface representation.

15 `surface` Commands

The `surface` command family works with triangle surfaces generated from voxels. These commands are deliberately separate from `voxel` commands because future DSMC-coupled runs will need to refresh surfaces, apply DSMC surface tallies, and then map that information back to voxels inside input-file loops.

15.1 Syntax

```
surf_flux <surface-id> source <source-id> select all
surf_flux <surface-id> source <source-id> select normal nx <x> ny <y> nz <z> [min-cos <c>]
surf_flux <surface-id> source <source-id> select voxels voxels <model>
surf_flux <surface-id> kinetic/theory pressure <p> temperature <T> \
  mole-fraction <x> molecular-mass <kg> reaction-prob <alpha> \
  solid-mass-per-hit <kg> select <selector>
surf_flux <surface-id> dsmc/surf fix <fix-id> quantity incident-number-flux \
  [reaction <path> | solid-mass-per-hit <kg>] \
  [reaction-prob <p>] [ablation-dt <dt> | mass-courant <C>]
surf_flux <surface-id> dsmc/surf fix <fix-id> column <N> \
  [reaction <path> | solid-mass-per-hit <kg>] \
  [reaction-prob <p>] [ablation-dt <dt> | mass-courant <C>]
surf_flux <surface-id> dsmc/reaction fix <fix-id> column <N> \
  sample-steps <Nstep> reaction <path> \
  [ablation-dt <dt> | mass-courant <C>] [time-scale <S>]
surf_measure_flux <surface-id> dsmc/reaction fix <fix-id> column <N> \
  sample-steps <Nstep> expected kinetic/theory number-density <n> \
  mole-fraction <x> temperature <T> molecular-mass <kg> \
  [reaction <path>]

surf_dump <dump-id> <surface-id> vtp <N> <path>
surf_dump off

# DSMC bridge only:
surf_write_vtp <path> <surface-id> [fix <fix-id> fields <name1> ...]
```

15.2 Examples

```
surf_flux skin source q1 select all
surf_flux skin source q1 select normal nx 1.0 ny 0.0 nz 0.0 min-cos 0.5
surf_flux skin source q1 select voxels voxels solid
surf_flux skin kinetic/theory pressure 50.0 temperature 5000.0 \
  mole-fraction 0.21 molecular-mass 5.313e-26 reaction-prob 1.0 \
  solid-mass-per-hit 1.3011869411625376e-23 select all
surf_flux skin dsmc/surf fix sflux mass-courant 0.25
surf_flux skin dsmc/reaction fix rco column 1 sample-steps 20 \
  reaction carbon-co.surf time-scale 1500
surf_measure_flux skin dsmc/reaction fix rco column 1 sample-steps 1 \
  expected kinetic/theory number-density 7.244e23 mole-fraction 0.21 \
  temperature 5000.0 molecular-mass 5.31352e-26 reaction carbon-co.surf

surf_dump skin skin vtp 10 output/sphere/surface_*.vtp
surf_dump off

surf_write_vtp skin output/surface-*.vtp \
  fix sqty fields collision-count incident-number-flux pressure
```

15.3 Description

`surf_flux` assigns one timestep of mass loss to selected triangles on an existing surface. Source and kinetic-theory modes require an explicit `select all`, `select normal`, or `select voxels` clause. With `source`, the source is a constant mass flux in `kg/m2/s`. The requested mass on each selected triangle is:

```
mass = flux * triangle-area * timestep
```

With `kinetic/theory`, the command computes the mass flux from ideal-gas impingement theory for one reactive species:

```
Gamma = mole-fraction * pressure/(kB*temperature)
        * sqrt(kB*temperature/(2*pi*molecular-mass))
flux = reaction-prob * solid-mass-per-hit * Gamma
```

This is intended as the first DSMC-coupled verification source: DSMC can run a spatially uniform, stationary hot gas domain while this command applies the corresponding continuum kinetic-theory flux to the current ISTHMUS triangles.

With `dsmc/surf`, the command reads per-surface data from a DSMC `fix ave/surf` instance. The preferred pattern is to average only the incident number flux:

```
compute sflux surf all gas nflux_incident
fix sflux ave/surf all 1 100 100 c_sflux[*] ave one
```

Then either select it by name with `quantity incident-number-flux`, or omit the quantity because it is the default for `dsmc/surf`. The bridge converts the selected number flux to solid mass flux with:

```
flux = number-flux * reaction-probability * solid-mass-per-hit
```

If neither `reaction` nor `solid-mass-per-hit` is supplied, IAC infers one solid formula unit per incident hit from the current voxel material. For a material defined as `formula C molar-mass 0.0120107`, this compact default consumes one carbon atom per counted incident O2 molecule. It is useful for kinetic-theory collision-flux examples where DSMC is not actually running surface chemistry.

When `reaction <path>` is used, IAC reads the same SPARTA `surf_react prob` file used by DSMC. For a file such as:

```
O2 --> CO + CO
D S 1.0
```

and a material defined as `formula C molar-mass 0.0120107`, IAC infers a reaction probability of `1.0` and a solid mass per hit equal to two carbon atoms. The raw `solid-mass-per-hit` and `reaction-prob` arguments are retained as overrides for cases whose solid consumption cannot be inferred from the material formula or gas-side reaction formula.

This is the first true DSMC-coupled source path. DSMC owns the gas domain, surface collisions, surface averaging, loops, and remapping commands; this library owns the voxel mass ledger, ISTHMUS surface ownership map, and ablation. The current bridge implementation supports this source on one MPI rank while the command and data path are kept compatible with later distributed storage.

With `dsmc/reaction`, the command reads per-surface reaction counts from a DSMC `fix ave/surf`, usually one that averages `compute react/surf`. The selected column is interpreted as an averaged number of simulation-particle reactions per surface element per sampled timestep. The bridge converts it to real solid mass removed over the coupling window with:

```
mass = reaction-count * sample-steps * fnum * solid-mass-per-reaction
```

and then divides by triangle area and the ablation timestep before passing the equivalent mass flux to the voxel ledger. This is the preferred path for chemically reacting DSMC ablation because SPARTA owns the surface reaction probability, species conversion, and reaction tallies. Prefer `reaction <path>` so the solid mass per reaction is inferred from the same reaction file; use `solid-mass-per-reaction` only as an explicit override.

`surf_measure_flux` reads the same kind of DSMC reaction count data but does not apply mass loss. It sums the sampled reactions over the installed surface, converts them to a number flux with:

```
reaction-flux = sum(reaction-count) * fnum / (surface-area * DSMC-timestep)
```

and stores scalar diagnostics that can be checked later with `iac verify`. The `expected kinetic/theory` arguments compute the ideal-gas one-way impingement flux for a reactive species:

```
expected-reaction-flux =
  reaction-prob * mole-fraction * number-density
  * sqrt(kB*temperature/(2*pi*molecular-mass))
```

The command also stores `reaction-count-per-step` and `expected-reaction-count-per-step`, which are often the clearest DSMC sampling diagnostics. If `reaction` or `solid-mass-per-reaction` is supplied, it additionally stores `reaction-mass-flux` and `expected-reaction-mass-flux` in kg/m2/s by multiplying the number flux by the inferred or supplied solid mass consumed per reaction.

This command is the first DSMC-hosted regression hook for the coupled path: it tests geometry generation, surface installation, DSMC surface chemistry tallies, core diagnostics, and input-file verification without introducing voxel deletion or remeshing.

By default, DSMC-coupled sources advance voxel ablation by the elapsed DSMC time since the previous coupling update. The optional `ablation-dt` argument overrides that physical ablation time. For `dsmc/reaction`, `time-scale` multiplies both the sampled reaction-count mass and the ablation time. This is useful for quasi-steady chemistry probes: the input can sample a short DSMC window and advance the solid over a longer reservoir-equivalent ablation time. When comparing multiple DSMC sampling lengths at the same ablation time, choose `time-scale = ablation-update-time / (sample-steps * timestep)`.

The optional `mass-courant` argument chooses the ablation timestep from the largest current local voxel-face flux. For DSMC surface fluxes, the limiting flux is the largest positive triangle mass flux that maps to an active, non-fixed voxel:

```
dt = C * density * voxel-size / max(triangle-flux)
```

This is the same nondimensional definition as standalone 'timestep mass/courant': it is the mass that the local flux would remove through one voxel face during the timestep divided by one voxel mass. Thus `mass-courant 0.1666666667` is the conservative value `1/6`; even if all six faces of a voxel saw that same limiting flux, the update cannot remove more than one voxel mass before carryover/deletion handles the remainder. `ablation-dt` and `mass-courant` are mutually exclusive.

The mass is stored on the triangle until a later command consumes it:

```
voxel_ablate solid surface skin policy local delete yes
```

Selectors:

- `select all` applies flux to every triangle.
- `select normal` applies flux only when the triangle normal has

`dot(normal, direction) >= min-cos`.

- `select voxels` applies flux to triangles that have ISTHMUS ownership

entries for the named voxel model. This is a forward-compatible hook for voxel groups and material subsets.

`surf_dump` writes VTP triangle files on scheduled run steps. The VTP cell data includes `area`, `requested-mass`, and `last-requested-mass`. The `last-requested-mass` field is useful for visual inspection because `requested-mass` is cleared after `voxel_ablate` consumes it.

Inside DSMC, scheduled `surf_dump` output is written when bridge commands advance the core step. Define the dump before the first surface generation so the active core model is initialized with the desired schedule.

`surf_dump off` clears surface dumps that were already defined. Regression wrappers can use it after including a visual example input.

Inside DSMC, `surf_write_vtp` writes a one-shot VTP snapshot immediately. This mirrors `voxel_write_vtu`: DSMC owns the run loop, and the input script can write the current ISTHMUS surface after each `isthmus_surf` regeneration. If the path does not contain `*`, the current core step number is inserted before the file extension.

The optional `fix <fix-id> fields <name1> ...` form reads per-surface values from a DSMC `fix ave/surf` and appends them as VTP triangle cell fields. The number of field names must match the number of per-surface columns exposed by the fix. This is the preferred visualization path for coupled DSMC runs because the ISTHMUS triangle geometry and DSMC surface quantities are written into the same VTP file.

15.4 Current Limits

- Only VTP surface dumps are implemented.
- Constant source flux and single-species kinetic-theory flux are implemented.
- DSMC `fix ave/surf` flux ingestion currently supports one MPI rank.
- `select voxels` currently distinguishes ownership presence for the current

voxel model, not separate voxel groups.

16 stats and stats_style Commands

The stats and stats_style commands control standalone console output.

16.1 Syntax

```
stats <N>
stats_style <column> <column> ...
```

16.2 Example

```
stats 1
stats_style step time active-voxels deleted-voxels remaining-mass mass-fraction volume-fraction front
```

16.3 Description

stats N prints every N steps and always prints step 0 and the final step.

stats_style selects columns from tracked history quantities. The standalone runner prints a title/configuration block first, then a fixed-width stats table.

16.4 Current Columns

```
step
time
active-voxels
deleted-voxels
remaining-mass
mass-fraction
volume-fraction
radius
front
requested-mass-step
applied-mass-step
dropped-mass-step
```

16.5 DSMC/SPARTA Note

In DSMC-hosted runs, use iac stats and iac stats_style for the core's own ablation table:

```
iac stats 1
iac stats_style step time active-voxels deleted-voxels remaining-mass mass-fraction radius
```

These commands do not change DSMC's native stats or stats_style settings. For coupled gas/solid runs, DSMC can continue printing particle/collision statistics while iac stats prints the voxel mass ledger after IAC ablation steps. Longer term, selected IAC quantities can also be exposed through DSMC compute/fix styles for native c_... or f_... output.

17 `voxel_dump` Command

The `voxel_dump` command writes voxel-model output.

Triangle surface dumps are configured with `surf_dump`.

17.1 Syntax

```
voxel_dump <id> <model> history <N> <path>
voxel_dump <id> <model> vtu <N> <path> [select all|active] [scalar <field>]
voxel_dump off

# Also available in the DSMC bridge:
voxel_dump <id> <model> history <N> <path>
voxel_dump <id> <model> vtu <N> <path> [select all|active] [scalar <field>]
voxel_write_vtu <model> <path> [select all|active] [scalar <field>]
```

17.2 Examples

```
voxel_dump hist solid history 1 output/slab-direct-ablation/history.csv
voxel_dump vox solid vtu 1 output/slab-direct-ablation/voxels_*.vtu select active scalar mass-fraction
voxel_dump off

voxel_write_vtu solid output/voxels-*.vtu select active scalar mass-fraction
```

17.3 Description

The `history` style writes one CSV summary after the run.

`voxel_dump off` clears voxel dumps that were already defined. This is useful in regression wrappers that include a visual example input but should avoid writing output files during CTest.

The `vtu` style writes VTK unstructured-grid files during the run. A dump is written at step 0, every `N` steps, and at the final step. If the path contains `*`, that character is replaced by a zero-padded step number:

```
output/slab-direct-ablation/voxels_000004.vtu
```

If the path does not contain `*`, the step number is inserted before the file extension.

The `select` option controls which voxels are written:

- `select all` writes every voxel, including inactive/deleted voxels.
- `select active` writes only active voxels.

The default is `select active`, so depleted voxels disappear from VTU output when the ablation fix uses `delete yes`.

The `scalar` option sets the active VTK cell scalar. This is the field ParaView should choose first when the file opens. Supported values are:

```
mass-fraction
remaining-mass
active
fixed
id
ix
```

```
iy
iz
```

The default is `scalar mass-fraction`.

Inside DSMC, scheduled `voxel_dump` output is written whenever a bridge command advances the core step, for example after `voxel_ablate`. This is useful for no-gas verification and coupled loops that use explicit DSMC input-file commands for ablation updates.

`voxel_write_vtu` writes a one-shot VTU snapshot immediately. This is useful in coupled loops where the input script decides exactly when a voxel snapshot should be written. It uses the same `select` and `scalar` options as scheduled `voxel_dump ... vtu` output. If the path does not contain `*`, the current core step number is inserted before the file extension.

17.4 History Columns

```
step,time,active-voxels,deleted-voxels,remaining-mass,mass-fraction,
volume-fraction,front,radius,requested-mass-step,applied-mass-step,
dropped-mass-step
```

17.5 VTU Cell Data

The VTU dump writes hexahedral voxel cells with:

```
mass-fraction
remaining-mass
id
ix
iy
iz
active
fixed
```

Use `active` for thresholding deleted voxels and `mass-fraction` or `remaining-mass` for coloring ablation progress in ParaView.

17.6 Planned Extensions

- Restart dumps.
- Material/field output.

18 `verify` Command

The `verify` command compares tracked quantities to input-file exact/reference expressions.

18.1 Syntax

```
verify <quantity> exact "<expression>" tolerance <value> [absolute|percent] norm <final|max|rms>
convergence <quantity> exact "<expression>" tolerance <value> [absolute|percent] norm <final|max|rms> vary <variable>
```

18.2 Examples

```
verify remaining-mass exact "initial-mass - q1*area*time" tolerance 0.01 percent norm max
verify mass-fraction exact "1.0 - q1*time/(rho*length)" tolerance 0.01 percent norm max
verify volume-fraction exact "1.0 - q1*time/(rho*length)" tolerance 0.01 percent norm final
verify front exact "q1*time/rho" tolerance 0.01 percent norm final
verify radius exact "initial-radius - q1*time/rho" tolerance 3.0 percent norm final
convergence radius exact "initial-radius - q1*time/rho" tolerance 100.0 percent norm final vary resolution 5 10 20 va
```

18.3 Description

The command evaluates an exact expression against the selected tracked quantity. The executable exits with a nonzero status if the error exceeds the tolerance. This makes verification cases natural CTest tests.

Regression tests generally use percent tolerances. Add `percent` after the tolerance value to check:

```
100 * abs(actual - exact) / abs(exact)
```

Percent error is undefined when the exact value is zero unless the actual value is also zero.

Use `absolute` or omit the mode when an absolute tolerance is more appropriate.

18.4 Convergence

`convergence` reruns the same input file with variable overrides and checks the error trend. Variables are defined with:

```
variable resolution equal 20
variable steps equal 4
```

and referenced with `${resolution}` or `${steps}` elsewhere in the input.

Each `vary` clause must have the same number of values. The first `vary` variable is treated as the refinement variable for apparent-order calculation. For example, `vary resolution 5 10 20 vary steps 1 2 4` runs three cases:

```
resolution=5,  steps=1
resolution=10, steps=2
resolution=20, steps=4
```

The convergence check requires monotone error reduction by default and checks that the apparent order from the first to last case is inside the requested `order <min> <max>` band.

18.5 Norms

`final` checks the last history row.

`max` checks the maximum absolute error over all recorded rows.

`rms` checks the root-mean-square error over all recorded rows.

18.6 Expression Variables

```
time
t
step
dt
rho
density
dx
nx
ny
nz
length
area
initial-mass
voxel-mass
active-voxels
deleted-voxels
remaining-mass
mass-fraction
volume-fraction
voxel-volume-fraction
front
q1
source:q1
```

19 `run` Command

The `run` command starts standalone time advancement.

19.1 Syntax

```
run <steps>
run duration <time>
```

19.2 Examples

```
run 8
run duration 0.02
```

19.3 Description

`run <steps>` advances a fixed number of steps.

`run duration <time>` advances for a requested physical duration using the current timestep. If the final step would overshoot, the model clips that step so the recorded time ends exactly at the requested duration.

For explicit ablation loops, use `limit time <target>` before `voxel_ablate` and `jump SELF <label> until time <target>` after `run 1`.

20 Direct Slab Ablation

This example removes exactly half of one voxel mass from each exposed slab column per step.

Input file:

```
units si

voxel_material carbon density 1800.0
voxel_create solid slab nx 8 ny 4 nz 4 dx 1.0e-6 material carbon

source q1 constant 1.8 units kg/m2/s
timestep mass/courant 0.5 source q1
variable ablation_time equal 4.0e-3

# Compact callback-style alternative:
# fix ab all voxel/ablate 1 voxels solid source q1 policy local face xlo delete yes

stats 1
stats_style step time active-voxels deleted-voxels remaining-mass mass-fraction volume-fraction front

voxel_dump hist solid history 1 output/slab-direct-ablation/history.csv
voxel_dump vox solid vtu 1 output/slab-direct-ablation/voxels_*.vtu select active scalar mass-fraction

label ablate-loop
limit time ${ablation_time}
voxel_ablate solid source q1 policy local face xlo delete yes
run 1
jump SELF ablate-loop until time ${ablation_time}
```

Run it:

```
./build/ia-core -in examples/slab-direct-ablation/in.slab-direct-ablation
```

The example writes:

```
output/slab-direct-ablation/history.csv
output/slab-direct-ablation/voxels_000000.vtu
output/slab-direct-ablation/voxels_000001.vtu
...
output/slab-direct-ablation/voxels_000008.vtu
```

The VTU dump uses `select active`, so voxels removed by `delete yes` disappear from the visualized voxel mesh. It also uses `scalar mass-fraction`, so ParaView should open the files with `mass-fraction` as the active cell field.

The test form is:

```
include ../../../examples/slab-direct-ablation/in.slab-direct-ablation

verify remaining-mass exact "initial-mass - q1*area*time" tolerance 0.01 percent norm max
verify mass-fraction exact "1.0 - q1*time/(rho*length)" tolerance 0.01 percent norm max
verify front exact "q1*time/rho" tolerance 0.01 percent norm final
```

Run the test:

```
ctest --test-dir build --output-on-failure
```

Print the stats table during the test:

```
cmake --build build --target check-verbose
```

Build the visual verification report:

```
cmake --build build --target test-report
```

The combined report is written to:

```
build/output/test-report.pdf
```

To report only this test, run:

```
python3 tools/run-test-report.py slab-direct-command-verification \
  --out build/output/slab-direct-report.tex
```

21 ISTHMUS Slab Ablation

These examples ablate an 8 by 4 by 4 slab from the `xlo` ISTHMUS surface.

The finite-patch case intentionally exposes lateral edges:

```
examples/slab-isthmus-ablation/in.slab-isthmus-finite
```

It runs:

```
isthmus_surf skin voxels solid buffer 1 weighting no map yes
surf_flux skin source q1 select normal nx -1.0 ny 0.0 nz 0.0 min-cos 0.5
voxel_ablate solid surface skin policy local delete yes
```

The ghost case adds y/z infinite-wall ghost voxels before surface generation:

```
voxel_ghost solid axis y boundary infinite layers 1
voxel_ghost solid axis z boundary infinite layers 1
isthmus_surf skin voxels solid buffer 1 weighting no map yes crop real
```

Ghost voxels are surface-construction images only. They do not carry separate mass, and triangle ownership maps back to the corresponding real voxels. This removes finite-patch edge effects while keeping the ablation update on the real voxel domain.

Run either case:

```
build/ia-core -in examples/slab-isthmus-ablation/in.slab-isthmus-finite
build/ia-core -in examples/slab-isthmus-ablation/in.slab-isthmus-ghost
```

The regression wrappers are:

```
tests/inputs/slab-isthmus-ablation/in.slab-isthmus-finite-surface.verify
tests/inputs/slab-isthmus-ablation/in.slab-isthmus-ghost-wall.verify
```

The finite-patch wrapper uses broad tolerances because the exposed edges are expected to perturb the one-dimensional exact recession. The ghost wrapper uses tight tolerances because the infinite-wall construction should recover the one-dimensional slab solution.

22 Sphere ISTHMUS Ablation

This example reconstructs a triangle surface from a voxel sphere with ISTHMUS, applies a constant mass flux to all surface triangles, maps the triangle mass loss back to voxels, and deletes depleted voxels.

The current case uses a 1 mm diameter sphere with `resolution 20`, meaning 20 voxels across the diameter. It uses `timestep mass/courant 0.5 source q1`, which gives `dt = 0.05625 s` for the current material, grid spacing, and flux. The four-step test therefore runs to `0.225 s`.

22.1 Inputs

```
examples/sphere-isthmus-ablation/in.sphere-isthmus-local
examples/sphere-isthmus-ablation/in.sphere-isthmus-normal
```

The core loop is:

```
isthmus_surf skin voxels solid buffer 1 weighting no map yes
surf_flux skin source q1 select all
voxel_ablate solid surface skin policy carryover/normal delete yes
run 1
```

This loop is intentionally command-driven. A future DSMC run can place DSMC convergence, flux mapping, voxel ablation, and surface regeneration in the same style of input-file loop.

22.2 Run

```
build/ia-core -in examples/sphere-isthmus-ablation/in.sphere-isthmus-local
build/ia-core -in examples/sphere-isthmus-ablation/in.sphere-isthmus-normal
```

22.3 Outputs

```
output/sphere-isthmus-local/
output/sphere-isthmus-normal/
```

Open the VTU files in ParaView to inspect active voxel mass fraction. Open the VTP files to inspect surface triangle area and the last applied surface mass request.

22.4 Verification

The automated wrappers are:

```
tests/inputs/sphere-isthmus-ablation/in.sphere-isthmus-local-deletion.verify
tests/inputs/sphere-isthmus-ablation/in.sphere-isthmus-normal-carryover.verify
```

It compares the inferred continuum-equivalent radius and mass fraction against the analytical shrinking-sphere solution:

```
radius = initial-radius - q1*time/rho
mass-fraction = (radius / initial-radius)^3
```

The local wrapper is the intentionally nonconservative reference path. It uses percent tolerances of `7.0 percent` for radius and `20.0 percent` for mass fraction at the final time.

The normal-carryover wrapper conserves mapped surface mass by pushing overshoot inward through live neighbor voxels. It uses tighter percent tolerances of `4.0 percent` for radius and `12.0 percent` for mass fraction, and it checks that final-step dropped mass is essentially zero.

23 DSMC Sphere Kinetic Example

This example runs from this repository using the private DSMC/IAC executable built by the overlay target. It is the simplest coupled DSMC recession case: SPARTA owns particles, surface collisions, `compute surf`, `fix ave/surf`, loops, `remove_surf`, and `surf_modify`; IAC owns voxel state, ISTHMUS surface generation, collision-flux-to-triangle conversion, normal-carryover ablation, and regenerated surface geometry.

Run it directly with:

```
cmake --preset dsmc
cmake --build --preset dsmc

cd examples/dsmc-sphere-kinetic
../../build-dsmc/bin/dsmc-iac \
  -screen none -log output/log.sparta -in in.dsmc-sphere-kinetic
```

To run the DSMC-enabled CTest suite:

```
ctest --preset dsmc
```

The normal standalone build does not include DSMC tests. A DSMC-enabled build runs the standalone tests plus DSMC-hosted bridge tests, including the `dsmc-sphere-kinetic-grid-convergence` check.

23.1 Collision-Flux Loop

The case uses pure O2 gas in a periodic 3 mm cube. Gas chemistry and gas-gas collisions are disabled on purpose. This makes the comparison a direct test of ideal-gas thermal impingement, DSMC surface-collision tallying, triangle-to- voxel mapping, normal carryover, and repeated ISTHMUS remeshing.

The gas and wall are both 5000 K:

```
boundary            p p p
species             air.species O2
mixture             gas O2 frac 1.0
mixture             gas temp 5000.0 vstream 0.0 0.0 0.0
surf_collide        1 diffuse 5000.0 1.0
```

SPARTA samples the incident number flux on each surface triangle:

```
compute             sflux surf all gas nflux_incident
fix                 sflux ave/surf all 1 100 100 c_sflux[*] ave one
run                 100 post no
```

IAC reads the incident number flux and converts it to carbon mass flux:

```
surf_flux skin dsmc/surf fix sflux mass-courant 0.1666666667
iac limit time ${ablation_time}
voxel_ablate solid surface skin policy carryover/normal delete yes
```

Because chemistry is intentionally disabled here, the command uses the compact `dsmc/surf` default: the incident flux is treated as a perfectly reactive collision flux and one carbon formula unit is consumed per counted O2 hit. With `voxel_material carbon ... molar-mass 0.0120107 formula C`, that means one carbon atom per hit. This is deliberately simpler than the reacting `surf_react` path; if the physical interpretation should be `O2 -> CO + CO`, the analytical comparison differs by a factor of two in solid mass consumed per incident O2 molecule. The DSMC-measured incident flux is still applied triangle-by-triangle, then mapped back to voxels through the ISTHMUS ownership fractions.

After each ablation update, native DSMC commands clear the old collision surface and install the regenerated ISTHMUS surface directly from memory:

```
remove_surf all
isthmus_surf skin voxels solid buffer 1 weighting no map yes
surf_install skin particle none type 1
surf_modify all collide 1
```

The committed example uses a 10-voxel sphere, `C_m = 1/6`, and a named `ablation_time` target. `iac limit time` clips the last solid timestep so the history ends at exactly that physical ablation time while exercising the full DSMC-to-ISTHMUS-to-voxel loop. It is a workflow example; the grid-refinement report below is the stronger accuracy check.

23.2 Analytical Comparison

For pure O2, the one-way thermal impingement flux is:

```
Gamma = n * sqrt(kB*T/(2*pi*m-O2))
j = Gamma * reaction-probability * solid-mass-per-hit
R(t) = R0 - j*t/rho-solid
mass-fraction = (R/R0)^3
```

The example writes:

```
examples/dsmc-sphere-kinetic/output/history.csv
```

Check its final mass fraction against the continuum solution with:

```
python3 ../../tools/check-dsmc-kinetic.py output/history.csv \
  --number-density 7.244e23 \
  --temperature 5000 \
  --molecular-mass 5.31352e-26 \
  --solid-density 1800 \
  --solid-molar-mass 0.0120107 \
  --solid-atoms-per-hit 1 \
  --initial-radius 5e-4
```

23.3 Grid-Refinement Report

The DSMC CTest suite runs this grid-refinement case without PDF generation. Build the visual DSMC grid-refinement report with:

```
cmake --build --preset dsmc --target dsmc-convergence-report
```

or run the generator directly:

```
python3 tools/run-dsmc-kinetic-convergence.py \
  --dsmc build-dsmc/bin/dsmc-iac \
  --resolutions 4,8,12 \
  --target-mass-fraction 0.2 \
  --sample-steps 10 \
  --mass-courant 0.3333333333 \
  --domain-half-width 6.5e-4 \
  --grid-cells 6 \
```

```
--fnum 2.0e11 \
--tolerance-percent 25 \
--require-improvement \
--pdf
```

The runner generates temporary input files under the build directory, computes the physical ablation time corresponding to the requested target mass fraction, and lets the runtime `mass-courant` command choose each solid ablation timestep from the current maximum triangle mass flux. The final update is clipped so each case ends at the same analytical ablation time. The pass/fail check uses the finest-grid mass fraction, voxelized volume fraction, and radius errors; `-require-improvement` also requires the finest mass error to improve over the coarsest case.

```
build-dsmc/output/dsmc-sphere-kinetic-grid-convergence/summary.csv
build-dsmc/output/dsmc-sphere-kinetic-grid-convergence/report.pdf
```

The report is optimized as a quick regression-style coupled check. It uses a small periodic box around the sphere, a modest `4,8,12` voxel-resolution ladder, and `C_m = 1/3` so the run remains short while still showing grid refinement toward the continuum sphere solution. The hand-run example above keeps the more conservative `C_m = 1/6` setting.

23.4 Chemistry Note

The repository still includes a one-step DSMC chemistry flux verification in:

```
tests/inputs/dsmc-sphere-flux/in.dsmc-sphere-flux.verify
```

That test keeps the `dsmc/reaction` bridge path covered. The main coupled sphere recession example intentionally uses collision flux instead, because it isolates the kinetic-theory comparison from chemistry, composition depletion, and heat-release modeling.

24 Directory Layout

```
include/isthmus_ablation/
  Public C++ headers for the core library.

src/
  Core implementation, parser, expression evaluator, and standalone CLI.

examples/
  Human-readable standalone input examples, organized as <case>/in.<case>.

dsmc-bridge/
  DSMC command-style bridge sources. The dsmc CMake target symlinks these into
  build-dsmc/dsmc-overlay/src; they should not be copied into the DSMC checkout.

tests/inputs/
  Regression wrappers, organized as <case>/in.<case>.verify.

docs/
  Manual, command reference, design notes, and generated diagram/PDF artifacts.

tools/
  Local documentation and development utilities.

build-dsmc/dsmc-overlay/
  Generated private DSMC source overlay. This is disposable build output.
```

Generated files are ignored:

```
build/
build-dsmc/
output/
site/
```

The generated single-file manual is:

```
docs/isthmus-ablation-core-manual.pdf
```

25 Build And Link

This project has two build modes:

- standalone `ia-core`, which runs only this repository's input commands;
- DSMC/IAC overlay, which builds a private DSMC executable with IAC commands.

The overlay is the recommended user workflow. It does not copy files into the user's DSMC checkout and it does not edit DSMC source files.

25.1 What Is Built

The core library is:

```
libisthmus_ablation_core.a
```

When ISTHMUS is found, that library links against:

```
libisthmus_cpp.a
```

The standalone executable is:

```
build/ia-core
```

The DSMC/IAC executable is:

```
build-dsmc/bin/dsmc-iac
```

25.2 Root Discovery

CMake looks for DSMC and ISTHMUS in this order:

- CMake cache variables:

```
cmake --preset dsmc -DDSMC_ROOT=/path/to/dsmc -DISTHMUS_ROOT=/path/to/isthmus
```

- Environment variables:

```
export DSMC_ROOT=/path/to/dsmc
export ISTHMUS_ROOT=/path/to/isthmus
```

- Common local paths:

```
../dsmc
~/dsmc
../isthmus/build
../isthmus/build-wsl
~/isthmus/build
~/isthmus/build-wsl
```

`ISTHMUS_ROOT` should point to an ISTHMUS checkout or install that contains a CMake package under `build/` or `build-wsl/`. Users can also set `ISTHMUS_CPP_DIR` or `isthmus_cpp_DIR` directly to the directory containing `isthmus_cppConfig.cmake`.

25.3 Recommended DSMC/IAC Build

One-time shell setup:

```
export DSMC_ROOT=$HOME/dsmc
export ISTHMUS_ROOT=$HOME/isthmus
```

Build and test:

```
cmake --preset dsmc
cmake --build --preset dsmc
ctest --preset dsmc
```

The preset file is intentionally kept at CMakePresets schema version 2 so it works with CMake 3.20 and newer. If a user has an older CMake, use the equivalent non-preset commands:

```
cmake -S . -B build-dsmc -DIAC_DSMC_USE_OVERLAY=ON
cmake --build build-dsmc --target dsmc
ctest --test-dir build-dsmc --output-on-failure
```

Run a coupled input:

```
build-dsmc/bin/dsmc-iac \
  -in examples/dsmc-sphere-kinetic/in.dsmc-sphere-kinetic
```

25.4 DSMC Machine Target

The default DSMC make target is:

```
mpi
```

Override it with:

```
cmake --preset dsmc -DDSMC_MACHINE=mac_mpi
cmake --build --preset dsmc
```

or:

```
cmake -S . -B build-dsmc -DIAC_DSMC_USE_OVERLAY=ON -DDSMC_MACHINE=mac_mpi
cmake --build build-dsmc --target dsmc
```

The executable generated by DSMC inside the overlay is `spa_<machine>`, and this repository creates a stable symlink:

```
build-dsmc/bin/dsmc-iac -> build-dsmc/dsmc-overlay/src/spa_<machine>
```

25.5 How The Overlay Works

The `dsmc` target creates:

```
build-dsmc/
  bin/
    dsmc-iac
  dsmc-overlay/
    README.md
    src/
      symlinks to DSMC src files
      symlinks to dsmc-bridge/*.cpp and *.h
      generated Makefile.package
      generated Makefile.package.settings
      Obj_<machine>/
      spa_<machine>
```

The script that creates this is:

```
tools/create-dsmc-overlay.py
```

It symlinks the user's DSMC `src` directory into the overlay, excluding build artifacts and generated style headers. It then symlinks these IAC bridge files:

```
dsmc-bridge/iac.cpp
dsmc-bridge/iac.h
dsmc-bridge/iacbridge.cpp
dsmc-bridge/iacbridge.h
dsmc-bridge/isthmus.cpp
dsmc-bridge/isthmus.h
dsmc-bridge/source.cpp
dsmc-bridge/source.h
dsmc-bridge/surface.cpp
dsmc-bridge/surface.h
dsmc-bridge/voxel.cpp
dsmc-bridge/voxel.h
```

The overlay also symlinks this repository's public include directory:

```
include/isthmus_ablation -> build-dsmc/dsmc-overlay/src/isthmus_ablation
```

This keeps bridge includes such as `#include "isthmus_ablation/model.hpp"` resolvable even during DSMC's dependency-generation rules, before the normal package include flags are applied consistently by every machine makefile.

DSMC's existing make system scans headers in the overlay and regenerates `style_command.h`. Because the bridge headers are present in the overlay, commands such as:

```
voxel ...
isthmus ...
surface ...
source ...
iac ...
```

become visible to DSMC without editing DSMC parser logic.

The overlay `Makefile.package` adds:

```
-I/path/to/isthmus-ablation-core/include
-I/path/to/isthmus/include
/path/to/isthmus-ablation-core/build-dsmc/libisthmus_ablation_core.a
/path/to/isthmus/build/libisthmus_cpp.a
```

or the equivalent paths discovered from the ISTHMUS CMake package.

25.6 Standalone Build

Standalone mode only needs this repository and, for ISTHMUS cases, ISTHMUS:

```
cmake --preset standalone
cmake --build --preset standalone
ctest --preset standalone
```

Run a standalone ISTHMUS example:

```
./build/ia-core -in examples/sphere-isthmus-ablation/in.sphere-isthmus-normal
```

If ISTHMUS is not found, CMake still builds direct voxel functionality, but ISTHMUS tests and examples are unavailable.

25.7 Debugging The Overlay

The overlay is disposable. To force a clean DSMC/IAC rebuild:

```
rm -rf build-dsmc/dsmc-overlay build-dsmc/bin/dsmc-iac
cmake --build --preset dsmc
```

To inspect the generated DSMC package settings:

```
cat build-dsmc/dsmc-overlay/src/Makefile.package
```

To bypass the overlay and run tests against an existing DSMC executable:

```
cmake -S . -B build-dsmc-external \
  -DIAC_DSMC_USE_OVERLAY=OFF \
  -DIAC_DSMC_EXECUTABLE=/path/to/spa_mac_mpi
cmake --build build-dsmc-external
ctest --test-dir build-dsmc-external --output-on-failure
```

That external path is mainly for debugging old builds. New users should prefer the overlay.

26 Testing

The project uses CTest.

Build and run tests:

```
cmake --preset standalone
cmake --build --preset standalone
ctest --preset standalone --output-on-failure
```

CTest captures output from passing tests by default. To see the stats table while the test runs, use verbose CTest output:

```
ctest --preset standalone --output-on-failure --verbose
```

or use the convenience target:

```
cmake --build --preset standalone --target check-verbose
```

To build the optional visual verification report:

```
cmake --build --preset report
```

This runs the tests, collects each configured test report CSV, and writes:

```
build/output/test-report.pdf
```

To select a subset directly:

```
python3 tools/run-test-report.py all
python3 tools/run-test-report.py 1-3
python3 tools/run-test-report.py slab-direct-command-verification
```

The standalone regression tests are:

```
slab-direct-command-verification
slab-direct-yhi-verification
slab-isthmus-finite-surface-verification
slab-isthmus-ghost-wall-verification
sphere-isthmus-local-deletion-verification
sphere-isthmus-normal-carryover-verification
sphere-isthmus-kinetic-theory-verification
sphere-isthmus-normal-carryover-convergence
```

These are standalone/core tests. They run with the `ia-core` binary and should be the only tests present in a normal standalone build.

To add coupled DSMC tests, configure and build the DSMC/IAC overlay:

```
cmake --preset dsmc
cmake --build --preset dsmc
ctest --preset dsmc
```

That build runs the standalone/core tests plus:

```
dsmc-slab-direct-ablation-verification
dsmc-sphere-isthmus-normal-carryover-verification
dsmc-sphere-flux-verification
dsmc-sphere-kinetic-grid-convergence
```

The DSMC tests are intentionally modest. The flux verification is a one-step instantaneous kinetic-theory check: it verifies the initial O2 reaction count and equivalent carbon mass flux before the perfectly consuming surface depletes the nearby O2 population. The kinetic grid-convergence test then runs the coupled collision-flux recession loop on a small `4,8,12` voxel-resolution ladder with `C_m = 1/3`. It compares final mass fraction, voxelized volume fraction, and radius to the analytical kinetic-theory sphere recession near 20% remaining mass, and requires the finest mass error to improve over the coarsest case. This test is only present in DSMC-enabled builds.

To build the DSMC convergence report:

```
cmake --build --preset dsmc --target dsmc-convergence-report
```

The target uses the same optimized `4,8,12` voxel-resolution ladder as the `dsmc-sphere-kinetic-grid-convergence` CTest, but also compiles the PDF report. The normal CTest path skips PDF generation so the pass/fail check stays quick.

It writes:

```
build-dsmc/output/dsmc-sphere-kinetic-grid-convergence/summary.csv
build-dsmc/output/dsmc-sphere-kinetic-grid-convergence/report.pdf
```

The standalone test report currently includes:

```
tests/inputs/slab-direct-ablation/in.slab-direct-command.verify
tests/inputs/slab-direct-ablation/in.slab-direct-yhi.verify
tests/inputs/slab-isthmus-ablation/in.slab-isthmus-finite-surface.verify
tests/inputs/slab-isthmus-ablation/in.slab-isthmus-ghost-wall.verify
tests/inputs/sphere-isthmus-ablation/in.sphere-isthmus-local-deletion.verify
tests/inputs/sphere-isthmus-ablation/in.sphere-isthmus-normal-carryover.verify
tests/inputs/sphere-isthmus-ablation/in.sphere-isthmus-kinetic-theory.verify
tests/inputs/sphere-isthmus-ablation/in.sphere-isthmus-normal-carryover-convergence.verify
```

The command-loop test includes the example case:

```
include ../../../examples/slab-direct-ablation/in.slab-direct-ablation
```

The fix test keeps the compact callback-style path covered while the examples move toward explicit `voxel_ablate` loops. Tests pass only if all `verify` commands in the wrapper input files pass.

The sphere ISTHMUS tests are enabled when the build finds the ISTHMUS C++ package. They run marching cubes, apply constant triangle flux, map that flux back to voxels, and compare the inferred radius and mass fraction against the continuum shrinking-sphere solution. The local sphere test keeps the nonconservative path covered. The normal-carryover sphere test checks the conservative path and verifies that final-step dropped mass is essentially zero.

The slab ISTHMUS tests compare a finite 4 by 4 surface patch against the same patch with y/z infinite-wall ghost voxels. The finite-patch case has broad tolerances because edge effects are expected. The ghost case should match the one-dimensional slab recession nearly exactly.

The convergence test is an input file, not an external runner. It uses `variable ... equal ...` and a `convergence ... vary ... order ...` command to run normal carryover at 5, 10, and 20 voxels across the diameter with mass-Courant timing. It requires monotone radius-error reduction. The voxelized volume-fraction check allows non-monotone coarse stair-step ties but still checks the end-to-end apparent order in a broad first-order band.

26.1 Test Organization

Use `tests/inputs/` for automated regression wrappers. Keep these compact and deterministic. Prefer including a file from `examples/` and adding only the `verify` commands or test-specific criteria in the wrapper.

Use `examples/` for readable user-facing examples. Examples should describe and run the physical case without embedding every regression criterion.

Example cases should use DSMC/SPARTA-style descriptive folders and input names:

```
examples/<case-name>/in.<case-name>
tests/inputs/<case-name>/in.<case-name>.verify
```

Standalone convergence tests should live under `tests/inputs/` and use the explicit `convergence` verification command. Coupled DSMC convergence tests can use a runner under `tools/` when they need to launch DSMC repeatedly and generate per-case input files under `build/output`.

27 Test Reports

Verification reports are optional visual artifacts for inspecting regression cases. They are not part of the core physics model.

27.1 Command-Line Report Output

The executable can write verification data when the test runner asks for it:

```
build/ia-core -in tests/inputs/slab-direct-ablation/in.slab-direct-command.verify \
  -report-csv build/output/test-report-data/slab-direct-command-verification/report.csv
```

The CSV contains:

```
quantity,expression,step,time,actual,exact,error,tolerance,tolerance-mode,norm,pass
```

This data-first design is intentional. Future DSMC/SPARTA-coupled runs can emit the same verification/report CSV without depending on standalone plotting code or modifying the input file.

27.2 Report Cases

Reportable tests are listed in:

```
tests/report-cases.csv
```

Each row gives the report index, test name, input file, and optional requirements. Add future regression cases there when they should be available to the report runner.

```
index,name,input,requires,expected-fail
2,slab-direct-yhi-verification,tests/inputs/slab-direct-ablation/in.slab-direct-yhi.verify,,no
3,slab-isthmus-finite-surface-verification,tests/inputs/slab-isthmus-ablation/in.slab-isthmus-finite-surface.verify,i
4,slab-isthmus-ghost-wall-verification,tests/inputs/slab-isthmus-ablation/in.slab-isthmus-ghost-wall.verify,isthmus,n
5,sphere-isthmus-local-deletion-verification,tests/inputs/sphere-isthmus-ablation/in.sphere-isthmus-local-deletion.ve
6,sphere-isthmus-normal-carryover-verification,tests/inputs/sphere-isthmus-ablation/in.sphere-isthmus-normal-carryove
7,sphere-isthmus-kinetic-theory-verification,tests/inputs/sphere-isthmus-ablation/in.sphere-isthmus-kinetic-theory.ve
8,sphere-isthmus-normal-carryover-convergence,tests/inputs/sphere-isthmus-ablation/in.sphere-isthmus-normal-carryover
```

The runner auto-detects optional features from the CMake build cache. Cases with unmet requirements are skipped when selecting `all`. Cases marked `expected-fail` are allowed to return a verification failure, and their CSV data is still included in the generated report.

27.3 Build The Combined Report

Run all report cases and build one combined PDF:

```
cmake --build build --target test-report
```

The generated files are:

```
build/output/test-report.tex
build/output/test-report.pdf
build/output/test-report-data/<test-name>/report.csv
```

The report includes:

- a table of contents for jumping to individual cases and input listings;
- a global summary table with case name, quantity, error, tolerance, norm, and pass/fail;
- one section per test case, starting on a new page;
- the test input listing for each case;
- any input files included by the test input, also starting on new pages;
- actual vs exact plots for each verified quantity;
- error plots with tolerance lines for each verified quantity;
- convergence order tables for convergence cases;
- convergence plots with each grid resolution shown in the legend.

27.4 Direct Tool Use

The report runner accepts the same selection styles we expect to use as the suite grows:

```
python3 tools/run-test-report.py all
python3 tools/run-test-report.py 1-4
python3 tools/run-test-report.py slab-direct-command-verification
python3 tools/run-test-report.py slab-direct-command-verification,slab-isthmus-ghost-wall-verification
```

Optional features can be overridden explicitly:

```
python3 tools/run-test-report.py all --available isthmus
python3 tools/run-test-report.py all --available ""
```

Names can be shortened if the partial name matches exactly one case:

```
python3 tools/run-test-report.py fix-verification \
  --out build/output/fix-report.tex
```

The lower-level report builder can still aggregate already-written CSV files:

```
python3 tools/build-test-report.py --discover build/output/test-report-data \
  --out build/output/discovered-report.tex --pdf
```

Discovery mode does not know input files unless named cases are provided, so the normal workflow uses `tools/run-test-report.py`.

27.5 Convergence Reports

When a test input uses the `convergence` command, the executable writes extra CSV files next to the normal `report.csv`:

```
convergence.csv
convergence-order.csv
convergence-<quantity>-<case>.csv
```

The report generator uses these files to add a convergence order table and plots. The table records the first and last refinement values, the corresponding errors, the measured order, the allowed order band, whether monotone error reduction was required, and the pass/fail result. The plots show the convergence error points and actual/exact time histories for each resolution with legend entries identifying the varied input values.

28 Documentation

Documentation is part of the development loop. Whenever a command or behavior is added or edited, update the relevant Markdown page in `docs/` in the same change.

28.1 Source Format

The manual source is plain Markdown. Command pages live in:

```
docs/commands/
```

Concept pages live in:

```
docs/concepts/
```

Examples live in:

```
docs/examples/
```

Development notes live in:

```
docs/development/
```

The web documentation layout is described by:

```
mkdocs.yml
```

The intended web-docs stack is MkDocs Material because it is modern, searchable, widely used, and still keeps every page as readable Markdown.

28.2 PDF Manual

The repository also has a local PDF builder:

```
cmake --preset standalone
cmake --build --preset docs
```

It writes:

```
docs/isthmus-ablation-core-manual.pdf
```

The lower-level builder is:

```
python3 tools/build-docs-pdf.py
```

This PDF builder is intentionally lightweight. It is a dependable review artifact for local development, while MkDocs Material remains the richer web documentation path.

28.3 Update Rule

When functionality changes:

- update the command reference for syntax changes;
- update concepts if the architecture or verification model changes;
- update examples when user-facing input files change;
- rebuild the PDF manual before handing the change back.